

02b - multiplexity

GR

2024-08-15

Load libraries

```
library(igraph)
library(tidyverse)
library(ggnetwork)
```

Multiplexity

- Lazega, Emmanuel. 2001. *The Collegial Phenomenon: The Social Mechanisms of Cooperation Among Peers in a Corporate Law Partnership*. Oxford University Press.
- <https://manliodedomenico.com/data.php>

```
lazega.df <- read_delim("Lazega-Law-Firm_multiplex.edges", delim=" ", col_names = F)
```

```
## Rows: 2571 Columns: 4
## -- Column specification -----
## Delimiter: " "
## dbl (4): X1, X2, X3, X4
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

```
colnames(lazega.df) <- c("layer", "ego", "alter", "weight")
lazega.df <- lazega.df %>%
  mutate(layer_str = case_when(layer==1 ~ "advice",
                                layer==2 ~ "friendship",
                                layer==3 ~ "co-work"))
glimpse(lazega.df)
```

```
## Rows: 2,571
## Columns: 5
## $ layer      <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ ego        <dbl> 1, 1, 1, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3, 3, 3, 3, 4, 4, 4, ~
## $ alter      <dbl> 2, 17, 20, 1, 6, 17, 20, 22, 24, 26, 2, 6, 14, 17, 18, 28, 3~
## $ weight     <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ layer_str  <chr> "advice", "advice", "advice", "advice", "advice", "advice", ~
```

```
table(lazega.df$layer_str)
```

```
##  
##      advice      co-work friendship  
##      892       1104       575
```

```
lazega.g <- lazega.df %>% select(ego,alter,layer_str,weight) %>% graph_from_data_frame(directed = F)  
edge.attributes(lazega.g) %>% table()
```

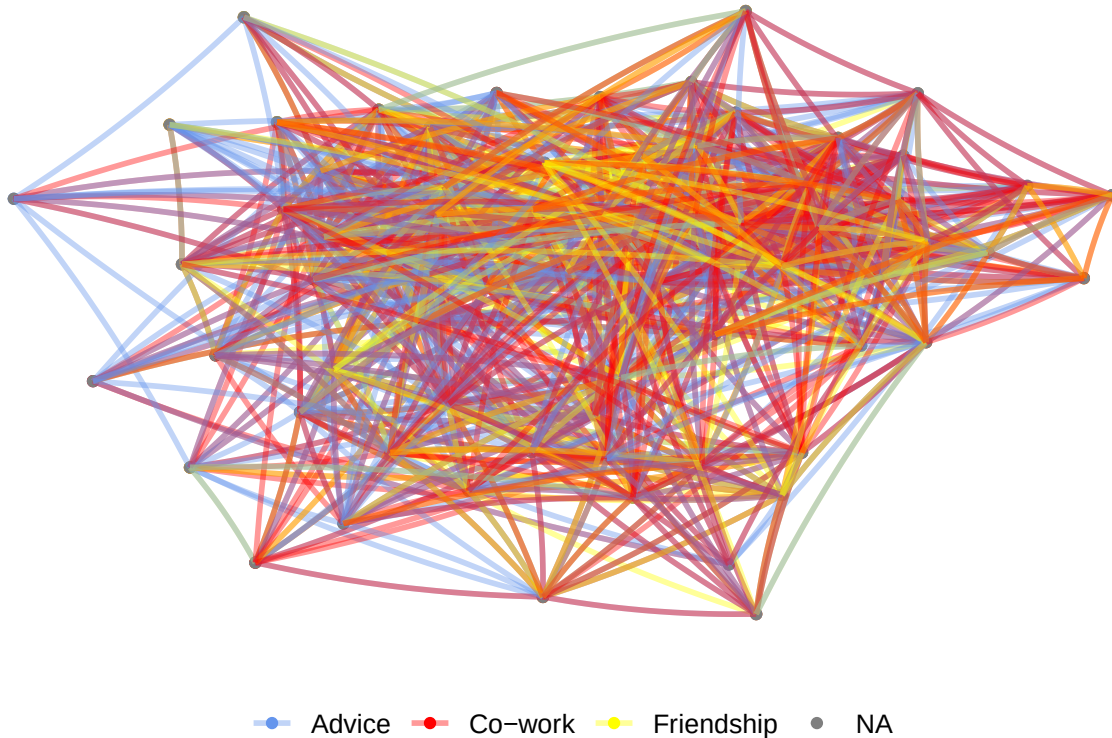
```
##           weight  
## layer_str      1  
##  advice      892  
## co-work     1104  
## friendship   575
```

```
gg.lazega <- ggnetwork(lazega.g, layout = with_fr(grid = "nogrid"))
```

```
## Warning in format_fortify(model = model, nodes = nodes, weights = "none", :  
## duplicated edges detected
```

```
gg.lazega %>%  
  ggplot(aes(x = x, y = y, xend = xend, yend = yend, color=as_factor(layer_str))) +  
  geom_nodes() +  
  geom_edges(lwd = 1,  
             alpha=0.4,  
             curvature = 0.05) +  
  scale_color_manual("", values = c("cornflowerblue", "red","yellow"),  
                     labels = c("Advice", "Co-work", "Friendship")) +  
  theme_blank() +  
  ggtitle("Lazega") +  
  theme(legend.position = "bottom", plot.title = element_text(hjust = 0.5))
```

Lazega



```
lazega.flat.df <- lazega.df %>% mutate(advice=if_else(layer==1,1,0),
                                     cowork=if_else(layer==2,1,0),
                                     friend=if_else(layer==3,1,0)) %>%
  select(ego,alter,advice,cowork,friend) %>%
  complete(ego,alter)
lazega.flat.df$advice[is.na(lazega.flat.df$advice)] <- 0
lazega.flat.df$cowork[is.na(lazega.flat.df$cowork)] <- 0
lazega.flat.df$friend[is.na(lazega.flat.df$friend)] <- 0

lazega.flat.df <- lazega.flat.df %>%
  group_by(ego,alter) %>%
  summarise(advice=sum(advice),cowork=sum(cowork),friend=sum(friend))
```

'summarise()' has grouped output by 'ego'. You can override using the '.groups' ## argument.

```
lazega.advice.cowork <- lazega.flat.df %>% xtabs(data = ., ~ advice + cowork)
lazega.advice.cowork
```

```
##      cowork
## advice    0    1
##      0 3932  217
##      1  534  358
```

```
lazega.advice.cowork %>% proportions(margin = 2)
```

```
##      cowork
## advice      0      1
##      0 0.8804299 0.3773913
##      1 0.1195701 0.6226087
```

```
lazega.advice.cowork %>% chisq.test()
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: .
## X-squared = 881.66, df = 1, p-value < 2.2e-16
```

```
lazega.flat.df %>% xtabs(data = ., ~ advice + friend) %>% chisq.test()
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: .
## X-squared = 1187.5, df = 1, p-value < 2.2e-16
```

```
lazega.flat.df %>% xtabs(data = ., ~ friend + cowork) %>% chisq.test()
```

```
##
## Pearson's Chi-squared test with Yates' continuity correction
##
## data: .
## X-squared = 361.89, df = 1, p-value < 2.2e-16
```

More complicated structures:

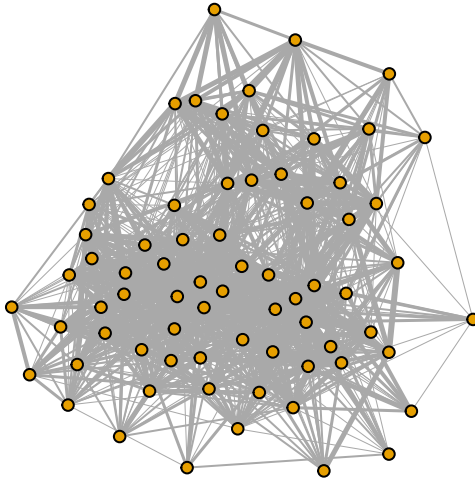
- multiplex triads. eg, Carmella's AddHealth hypothesis.
- Jacob G. Foster, Andrey Rzhetsky, James A. Evans. 2015. ASR. <https://arxiv.org/pdf/1302.6906>

Weight

```
lazega.weighted.g <- lazega.g %>%
  igraph::simplify(edge.attr.comb = list(weight="sum", "ignore"))
edge_attr(lazega.weighted.g)$weight %>% table()
```

```
## .
## 1 2 3 4 5 6
## 305 240 238 106 66 53
```

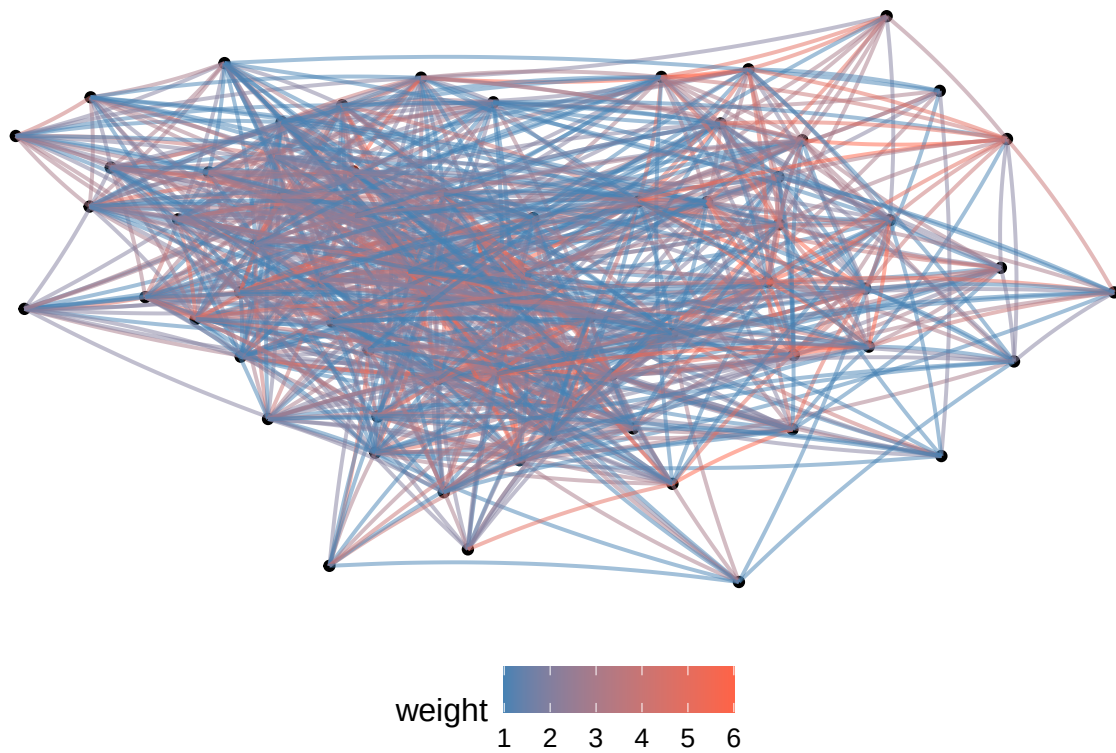
```
plot.igraph(lazega.weighted.g,
            vertex.label=NA,
            vertex.size=5,
            edge.width=E(lazega.weighted.g)$weight/2)
```



```
gg.lazega.weighted <- ggnetwork(lazega.weighted.g, layout = with_fr(grid = "nogrid"))

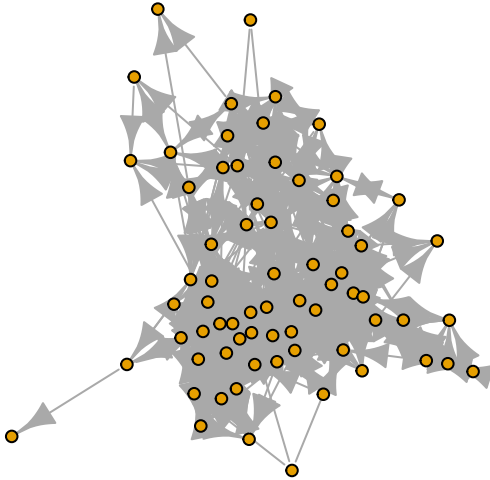
gg.lazega.weighted %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_nodes() +
  geom_edges(aes(color=weight),
             lwd = 0.7,
             alpha=0.5,
             curvature = 0.05) +
  scale_colour_gradient(low = "steelblue", high = "tomato") +
  theme_blank() +
  ggtitle("Lazega") +
  theme(legend.position = "bottom", plot.title = element_text(hjust = 0.5))
```

Lazega



Directed

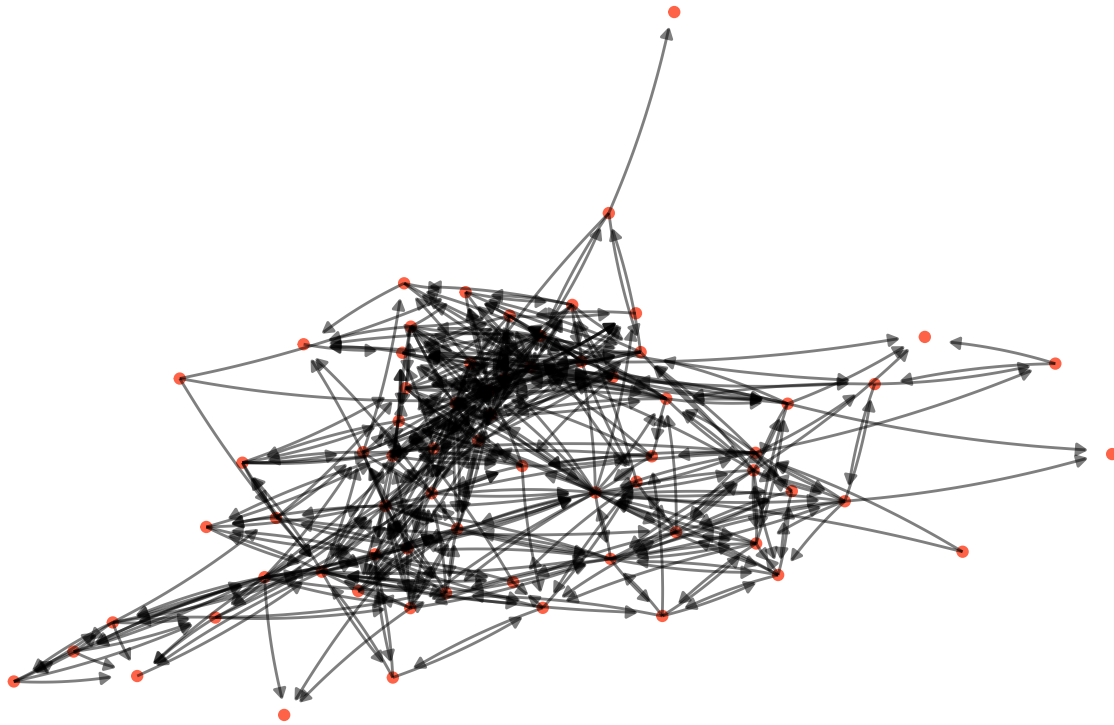
```
lazega.directed.g <- lazega.df %>% filter(layer_str=="friendship") %>%  
  select(ego,alter,layer_str,weight) %>%  
  graph_from_data_frame(directed = T)  
  
plot.igraph(lazega.directed.g,  
  vertex.label=NA,  
  vertex.size=5,  
)
```



```
gg.lazega.directed <- ggnetwork(lazega.directed.g, layout = with_fr(grid = "nogrid"))

# https://cran.r-project.org/web/packages/ggnetwork/vignettes/ggnetwork.html
gg.lazega.directed %>%
  ggplot(aes(x = x, y = y, xend = xend, yend = yend)) +
  geom_nodes(color="tomato") +
  geom_edges(arrow = arrow(length = unit(4, "pt"), type = "closed"), curvature = 0.05, alpha=0.5) +
  theme_blank() +
  ggtitle("Lazega") +
  theme(legend.position = "bottom", plot.title = element_text(hjust = 0.5))
```

Lazega



```
reciprocity(lazega.directed.g)
```

```
## [1] 0.6121739
```

```
shortest_paths(lazega.directed.g, 1, 60)
```

```
## $vpath
## $vpath[[1]]
## + 5/69 vertices, named, from 3bc9f47:
## [1] 1 17 43 64 66
##
##
## $epath
## NULL
##
## $predecessors
## NULL
##
## $inbound_edges
## NULL
```

```
shortest_paths(lazega.directed.g, 60, 1)
```

```
## $vpath
```

```
## $vpath[[1]]
## + 3/69 vertices, named, from 3bc9f47:
## [1] 66 39 1
##
##
## $epath
## NULL
##
## $predecessors
## NULL
##
## $inbound_edges
## NULL
```

```
mean_distance(lazega.directed.g, directed = T)
```

```
## [1] 2.505126
```

```
mean_distance(lazega.directed.g, directed = F)
```

```
## [1] 2.186275
```

```
lazega.distances <- distances(lazega.directed.g, mode="out")
```